Case E

Mr. Liam O'Connor, 19M — Cystic Fibrosis (Pulmonary Tune-up) | FEV■ decline · Increased purulent sputum volume & viscosity · Minor haemoptysis · Home PEP + exercise therapy · Poor adherence (university)

Cardiorespiratory Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced exercise capacity / deconditioning
- Unable to complete home PEP without exhaustion
- Exercise tolerance reduced (dyspnoea-limited)
- Difficulty maintaining therapy around university schedule

■■ RED FLAGS

- Significance
- Action if Present
- Frank Haemoptysis (>240 mL/24 h)
- Bronchial artery erosion; life-threatening airway haemorrhage risk
- Cease all airway clearance; urgent medical escalation; lateral decubitus (bleeding side down)
- SpO■ <88% at rest or with activity
- Severe hypoxaemia; risk of cor pulmonale and acute deterioration
- Halt exercise; apply supplemental O■; notify medical team; do not proceed without review

■ PATHOPHYSIOLOGY (KEY POINTS)

- Core defect: Mutations in the CFTR gene produce dysfunctional or absent chloride channels on airway epithelium.
- This leads to dehydrated, hyperviscous airway surface liquid — mucus that cannot be cleared by normal mucociliary action.
- Infection-inflammation cycle: Retained mucus plugs obstruct small and medium airways, creating a nidus for chronic infection — predominantly Pseudomonas aeruginosa .
- Neutrophil-dominated inflammation releases elastase and proteases that destroy airway walls, driving bronchiectasis, air trapping, and progressive obstructive lung disease.
- FEV■ decline mechanism: Each exacerbation accelerates bronchiectatic remodelling, reducing elastic recoil and increasing resistance.
- The cycle of infection → inflammation → structural damage compounds over time.

■ DIFFERENTIAL DIAGNOSES

1. Liam has a confirmed CF diagnosis with established CFTR dysfunction and

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observations

RR: target ≤25/min at rest SpO■: ≥93% on air (exacerbation baseline) Accessory muscle use → ↑ WOB Sputum: volume, colour, consistency Chest shape: barrel / hyperinflation

Auscultation Findings Expected

Widespread coarse crackles bilaterally — retained secretions in bronchiectatic airways Expiratory wheeze — dynamic airway collapse and obstruction Reduced air entry in lower zones — mucus plugging and air trapping

Spirometry — FEV■ / FVC (Pre & Post Bronchodilator)

What it measures Degree of airflow obstruction; tracks decline from personal best; assesses bronchodilator reversibility How to perform Patient seated, noseclip applied. Maximal inspiration then rapid, forceful,

OPEP Technique Assessment (Aerobika / Acapella)

What it measures Accuracy of current home airway clearance technique; identifies technique errors contributing to inadequate secretion mobilisation How to perform

■ TREATMENT PLAN

- therapy, mucolytics, recurrent tune-up admissions — is CF-specific.
- Non-CF bronchiectasis does not account for his CFTR genotype, the
- viscosity profile of his sputum, or his documented home PEP regimen.
- Test 2: Six-Minute Walk Test (6MWT) or Modified Shuttle Walk Test
- What it measures
- Functional exercise capacity; oxygen desaturation on exertion; baseline for exercise rehabilitation
- metre flat corridor (6MWT) or pre-recorded beep levels (MSWT). Monitor
- SpO■, HR, and Borg (0–10) at each minute. Record distance achieved. Stop
- Positive finding
- predicted 6MWT distance = impaired functional capacity. SpO■ drop
- >4% with exertion = exercise-induced hypoxaemia requiring O■ review.
- Why I chose this